Predictive Analytics for Road Safety

Predictive Analytics & Forecasting

Executive Master in Business Analytics and AI

Project 2026

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Description

You are part of an analytics team supporting a national road-safety authority. The organization wants to understand how historical highway-accident data can be used to support better road-safety decisions.

Highway Accidents Dataset

The dataset contains historical records of highway accidents in Brazil between 2007 and 2025. Each observation corresponds to an individual accident described by 23 variables covering the date and location of the accident, road characteristics and infrastructure, and accident severity.

For this project, you are provided with a sample of 150,000 accident records, preserving the temporal structure and main characteristics of the original dataset. Table 1 presents the data dictionary, with a description of each variable.

Tasks

Using the data set provided in **highway_accidents.csv**, you have a set of main tasks to accomplish as described next. Still, you are free to include other tasks to increase the value of your assignment.

Task 1: Problem Definition

The specific prediction task is intentionally not defined in advance. Based on the available data and the road-safety context described above, define a predictive problem that you consider relevant for supporting decision-making.

Clearly describe and justify:

- the business question you aim to address and the decision(s) your analysis could support;
- the prediction target and, when relevant, the unit of analysis and prediction horizon;
- how you formulate the problem as a predictive analytics task.

Your choices should be motivated by both the business objective and the information available in the dataset.

Table 1: Data Dictionary for highway-accidents dataset.

Variable	Description
date	Date of the accident.
km	Distance (km) from the beginning of the highway.
road_direction	Traffic-flow direction (increasing/decreasing km).
roadway_type	Road configuration (single/double/multiple).
area_type	Surrounding area (urban/rural).
latitude	Latitude of the accident location.
longitude	Longitude of the accident location.
time_of_day	Period of occurrence (early morning/morning/afternoon/night).
highway_type	Official highway classification.
Uphill	Presence of an uphill road segment.
Downhill	Presence of a downhill road segment.
Curve	Presence of a curve.
Straight	Presence of a straight road segment.
Crossing	Presence of a crossing.
Road_Intersection	Presence of a road intersection.
Roundabout	Presence of a roundabout.
Regulated_Uturn	Presence of a regulated U-turn.
Bridge	Presence of a bridge.
Tunnel	Presence of a tunnel.
Overpass	Presence of an overpass.
Temporary_Detour	Presence of a temporary detour.
Roadworks	Presence of roadworks.
accident_classification	Accident severity (no injuries / injuries / fatalities).

Task 2: Data Understanding and Preparation

Explore the data in relation to your question. Identify the information available, target distribution, missing values, unusual observations, potentially irrelevant variables and possible data leakage. Decide whether transformations or aggregations are required. Pay particular attention to the temporal dimension and to whether information would really have been available at prediction time.

Task 3: Predictive Modelling

From the available data, you should define the data set used for the predictive task at hand. Establish a simple baseline and compare it with at least two appropriate predictive approaches. Select metrics that are meaningful for the business question and design an evaluation strategy that reflects which solution would be used. Translate your results into actionable insight.

Tools

You can use R or Python. You can find material for dynamic reporting in R with markdown if you choose to use R. You can use the Colab Research Notebooks or Jupyter Notebooks if you decide to use Python.

Deliverables

The project is mandatory, should be performed in groups of two students, and submitted on moodle with a compressed file containing the following items:

- **slides for presentation (PDF format)** focusing on the main issues of the project for a 10 minutes presentation; any additional information that cannot be presented in the 10 min slot can be included as annexes to the presentation;

- the source of **two ready-to-execute dynamic reports or notebooks** with all the code necessary to reproduce the results you present for Task 2 and Task 3, i.e. it should be possible to replicate your predictions; any complementary files needed to execute your report (e.g. data files, data objects) should be submitted as well.

Deadlines

- Presentation: October 20th, 2026
- Submission: October 27th, 2026

Presentation Guidelines

- problem definition
- data understanding: concise summary, focusing on the main findings
- data preparation: outline, focusing on the main operations
- predictive modelling with experimental results
- conclusions, limitations and future work
- annexes

- TOTAL: max 40 slides

Grading

The grading of the practical assignment is distributed as follows:

- Task 1 (10%)
- Task 2 (15%)
- Task 3 (25%)
- Originality (10%)
- Presentation and discussion (40%) - **mandatory**